

A Study on Radiation Hardness of BGO Crystals

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Abstract— We report an investigation on the radiation hardness of 20 cm long BGO crystals produced at Shanghai Institute of Ceramics. Their optical and scintillation properties, such as transmittance and emission spectra, light output and response uniformity, were measured before, during and after γ -ray irradiations at 2, 8, 30 and 5,444 rad/h. BGO crystals produced in 2004 and 2011 are compared to a sample grown in early eighties for the L3 experiment at LEP. Significantly better radiation harness is observed in the sample grown in 2011 in transmittance, light output and radiation hardness. Their use for crystal calorimeters in future high energy physics experiments is discussed. To maintain crystal resolution, a precision light monitoring system is mandatory to trace variations of crystal transparency *in situ* for a BGO crystal calorimeter in a radiation environment. The origin of radiation damage in BGO is also discussed.

I. INTRODUCTION

BECAUSE of its high density (7.13 g/cm³), short radiation length (1.12 cm) and high light output (21% of NaI:TI), Bi₄Ge₃O₁₂ (BGO) crystals have been used to construct electromagnetic calorimeters (ECAL) for high energy physics (HEP) experiments, such as the L3 experiment at LEP [1]. It has also been considered for other experiments, where precision measurements of photons and electrons are important for their physics mission. The use of large size scintillation crystals in future HEP experiments, however, faces a challenge from the severe radiation environment expected in these experiments. In this paper, we present a study of the radiation hardness for large size BGO crystals grown at Shanghai Institute of Ceramics (SIC) by a modified Bridgman technique, and compare them to one sample grown in early eighties for the L3 experiment.

Fig.1 is a photo showing the BGO samples investigated in this study. The two rectangular samples were grown in 2004 and 2011 at SIC, and have a dimension of 25×25×200 mm³. The tapered BGO sample was grown at SIC in early eighties for L3 with a dimension of 20×20 mm² at the small end, 30×30 mm² at the large end and 240 mm in length. All three BGO crystals were annealed at 200°C for 200 minutes and then irradiated by γ -rays at 2, 8, and 30 rad/h step by step until reaching equilibrium, following which recovery was measured. The sample SIC-2011 was also irradiated under a high dose rate at 5,444 rad/h.

Longitudinal transmittance (LT) was measured using a PerkinElmer LAMBDA 950 UV/Vis spectrophotometer with double beam, double monochromator and a large sample

compartment equipped with an integrating sphere. The systematic uncertainty in repeated measurements was about 0.15%. Scintillation light output (LO) was measured before and after irradiations by using a Hamamatsu R1306 PMT with a bi-alkali photo-cathode and a borosilicate glass window. For the LO measurement, the large (tapered samples) or one (rectangular samples) end of the samples was coupled to the PMT with Dow Corning 200 fluid, while all other faces of the sample were wrapped with Tyvek paper. A collimated ¹³⁷Cs source was used to excite the samples. The γ -ray peak positions were obtained by a simple Gaussian fit. The γ -ray irradiations were carried out in two facilities at Caltech: a 50 curie ⁶⁰Co source for irradiations at low dose rates (2, 8 and 30 rad/h) and a 7,000 curie ¹³⁷Cs source for a high dose rate at 5,444 rad/h. The dose rates were calibrated.

The main consequence of γ -ray induced radiation damage in BGO crystals is radiation induced absorption which causes degradation of transmittance and light output [2, 3]. It may also affect crystal's light response uniformity (LRU) and cause a degradation of the energy resolution if the light attenuation length (LAL) is reduced significantly. The γ -ray induced absorption may also spontaneously recover, leading to dose rate dependent radiation damage [4, 5].



Fig. 1. BGO samples used in this study.

II. INITIAL PROPERTIES

Fig.2 shows the longitudinal transmittance and the emission spectra for all samples before irradiations. Also shown in the figure is the numerical value of the emission weighted longitudinal transmittance (EWLT), which is defined as:

$$EWLT = \frac{\int LT(\lambda) Em(\lambda) d\lambda}{\int Em(\lambda) d\lambda} \quad (1)$$

The EWLT is a good representation of crystal's transparency better than the transmittance at the emission peak. Comparing with other BGO samples, sample SIC-2011 shows the best LT

Manuscript received November 18, 2012. This work was supported in part by the U.S. Department of Energy under Grant No. DE-FG03-92-ER40701.

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and the highest EWLTL, indicating a clear progress in crystal's optical quality.

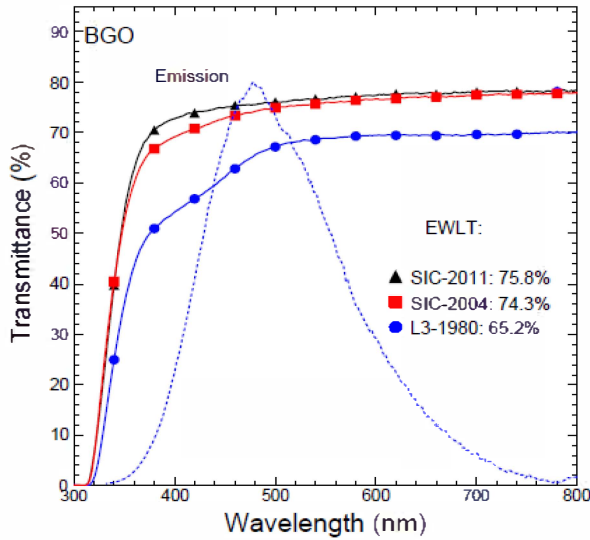


Fig. 2. Initial LT and EWLTL of BGO samples.

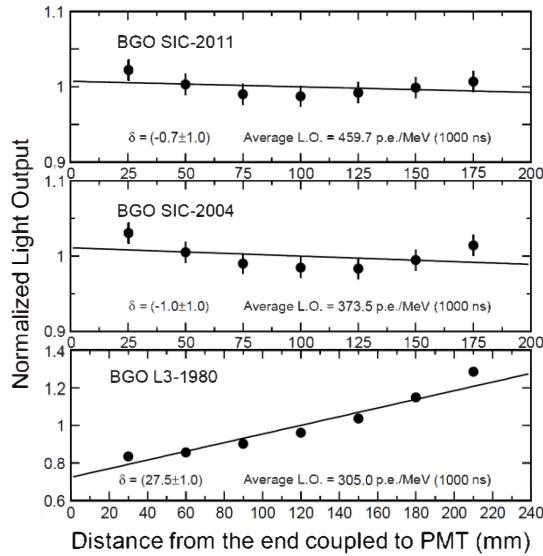


Fig. 3. Initial LO and LRU of BGO samples.

Fig. 3 shows the LO and LRU before irradiations for all samples. The LRU was measured by moving a collimated γ -ray source along the longitudinal axis of the long sample at seven evenly distributed points. The response (y) was fit to a linear function:

$$\frac{y}{y_{mid}} = 1 + \delta \left(\frac{x}{x_{mid}} - 1 \right) \quad (2)$$

where y_{mid} represents the light output at the middle of the crystal, δ represents the slope of the LRU; x is the distance from the crystal end which is coupled to the PMT. It is clear that sample SIC-2011 has the best LO among three samples. While both rectangular crystals have a flat (uniform) response, the response of the L3 BGO at the far (small) end is

significantly higher than that at the near (large) end which is coupled to the PMT. This is due to a well-known optical focusing effect caused by the tapered geometry of this sample.

III. GAMMA-RAY INDUCED RADIATION DAMAGE AND ITS RECOVERY

The longitudinal transmittances and EWLTLs before irradiation and at equilibrium under each dose rate are shown in Figs. 4 to 6 for the three samples respectively. The corresponding LRUs are shown in Figs. 7 to 9. It is clear that the samples SIC-2011 has also a better initial light output. We also note that the LRU of SIC-2011 is more or less maintained up to 5,444 rad/h, while the others show a significant degradations under much low dose rates (8 and 2 rad/h).

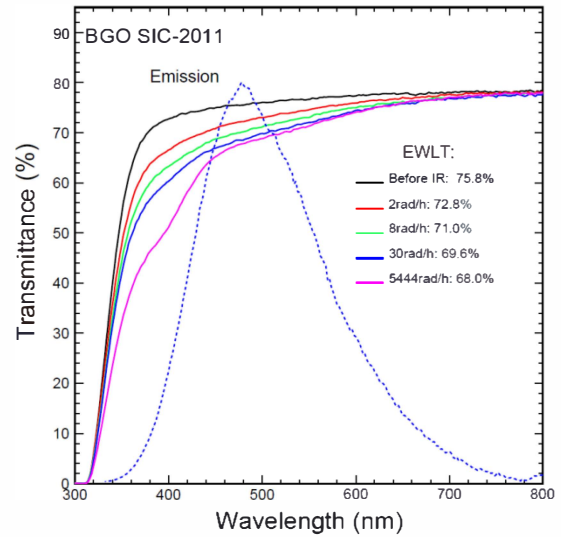


Fig. 4. The longitudinal transmittances and EWLTL values are shown for SIC-2011 before irradiation and at equilibrium under each dose rate.

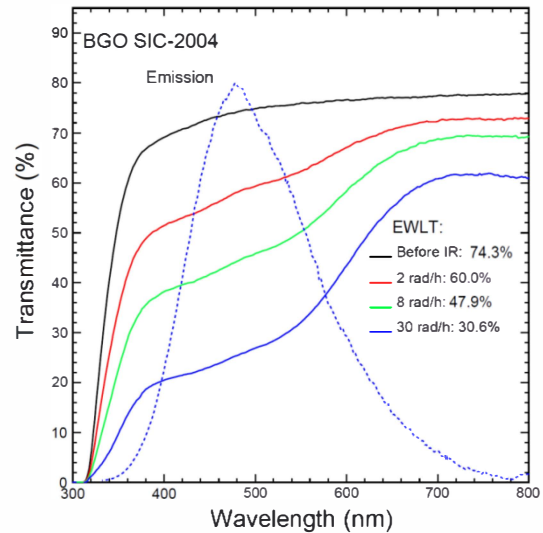


Fig. 5. The longitudinal transmittances and EWLTL values are shown for SIC-2004 before irradiation and at equilibrium under each dose rate.

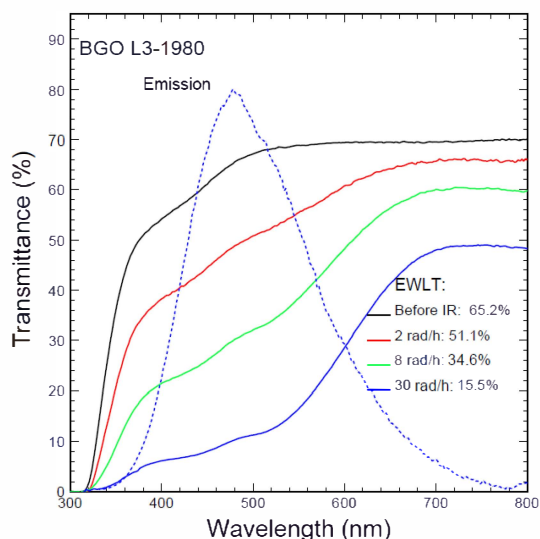


Fig. 6. The longitudinal transmittances and EWL values are shown for L3-1980 before irradiation and at equilibrium under each dose rate.

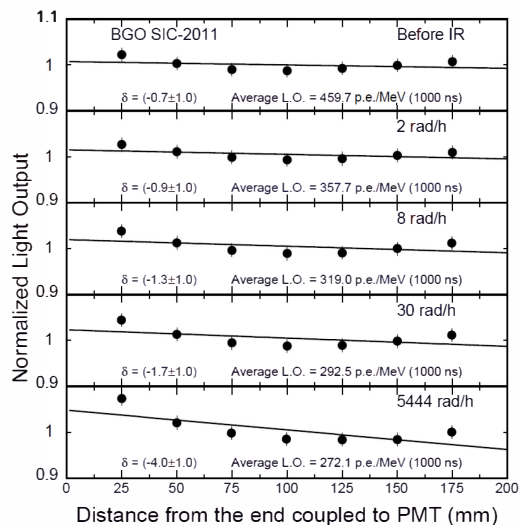


Fig. 7. The light response uniformities are shown for SIC-2011 before irradiation and at equilibrium under each dose rate.

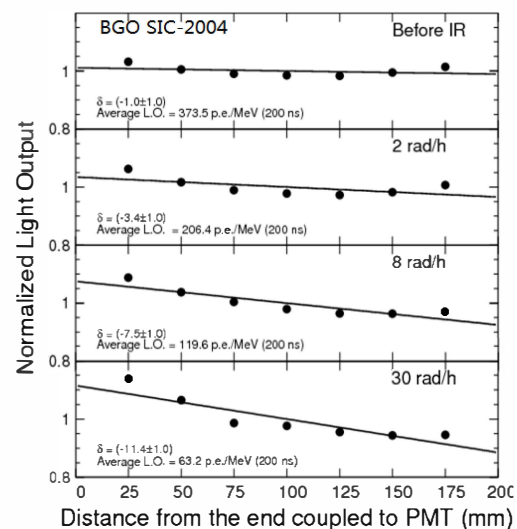


Fig. 8. The light response uniformities are shown for SIC-2004 before irradiation and at equilibrium under each dose rate.

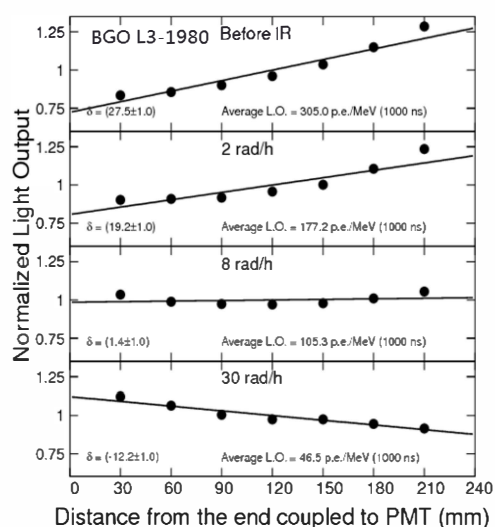


Fig. 9. The light response uniformities are shown for L3-1980 before irradiation and at equilibrium under each dose rate.

Fig. 10 shows that γ -ray induced radiation damage (EWLT) recovers under room temperature. Two time constants were determined by an exponential fit to describe the recovery, indicating at least two color centers at different depth. These two time constants are in an order of several days and several weeks which are in a good agreement with previous measurements [6]. The EWLT values measured after a few thousands hours are significant lower than the initial values (75.8%, 74.3% and 65.2% respectively for SIC-2011, SIC-2004 and L3-1980), indicating additional deep color centers in these crystals.

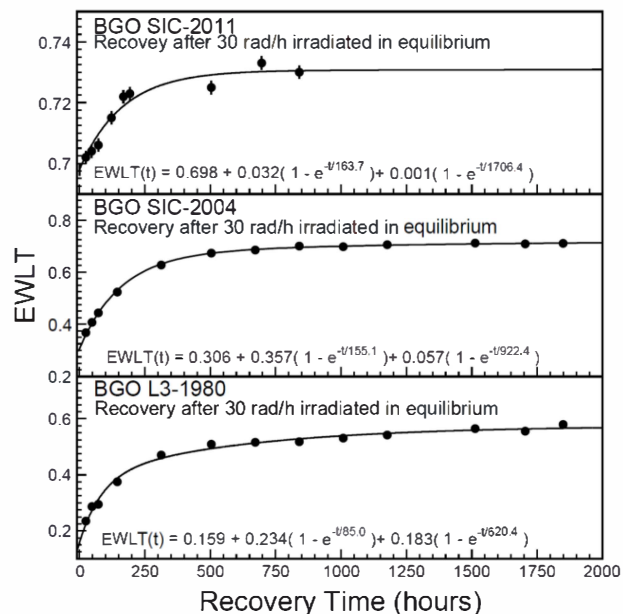


Fig. 10. Recoveries of EWLT at reaching equilibrium at 30 rad/h are shown for all three BGO samples.

Because of the recovery, the γ -ray induced radiation damage in BGO is dose rate dependent. The degradation of both LT and LO under a certain dose rate would reach an

equilibrium, when the speed of the color center formation (damage) equals to the speed of the color center annihilation (recovery) [4, 5]. Fig. 11 shows the normalized LO and EWLTL for SIC-2011 during γ -ray irradiations in four steps at dose rates of 2, 8, 30 and 5,444 rad/h and corresponding exponential fits. Both EWLTL and LO reached equilibrium during irradiations under a defined dose rate, showing a clear dose rate dependent damage. Consistent time constants were observed for both normalized LO and EWLTL.

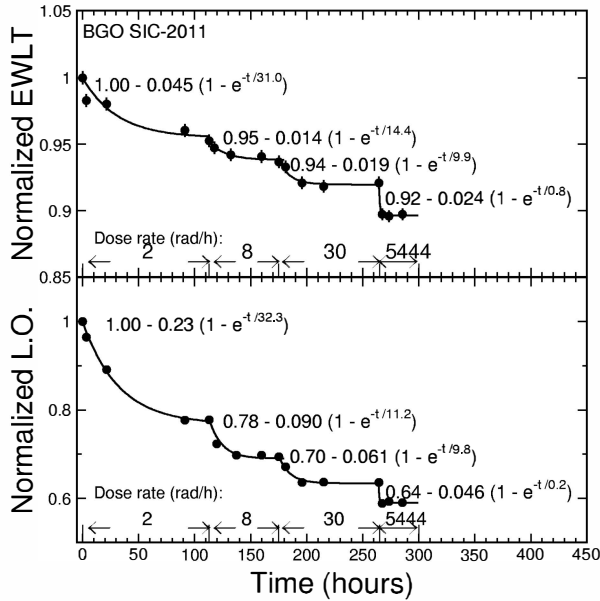


Fig. 11. Normalized LO and EWLTL are shown for SIC-2011 under γ -ray irradiations at different dose rates.

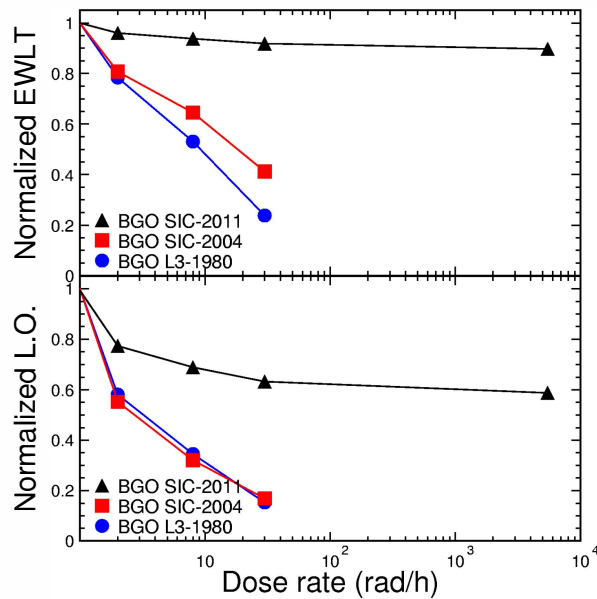


Fig. 12. Normalized LO and EWLTL are shown as a function of dose rate for all three BGO samples.

Fig. 12 shows the normalized LO and EWLTL as a function of the dose rate for all three samples in the equilibrium. The corresponding data are listed in Table 1. It is clear that the

radiation hardness of the sample SIC-2011 is the best among all three samples, while the other two samples have similar radiation hardness, indicating that the sample SIC-2011 has a much low defect density than other two samples produced early. This observation also indicates that radiation hard BGO crystals may be developed by reducing crystal lattice structure related defects, e.g. oxygen vacancies, through purity control of raw materials and refining growth parameters [2]. BGO crystals of quality similar to SIC-2011 may find application in future high energy physics experiments where severe radiation is expected. One intrinsic drawback of BGO application is its slow decay time of 300 ns, which is not adequate for applications requiring fast counting rate.

Table I. A Summary of LO and EWLTL losses

BGO Samples	SIC-2011	SIC-2004	L3-1980
Initial L.O. (p.e./MeV)	460	373	305
Initial EWLTL (%)	75.8	74.3	65.2
L.O. Loss (%)	2 rad/h	22	42
	8 rad/h	31	66
	30 rad/h	36	85
	5,444 rad/h	41	
EWLTL Loss (%)	2 rad/h	4	22
	8 rad/h	6	47
	30 rad/h	8	76
	5,444 rad/h	10.3	

IV. SUMMARY

The BGO sample SIC-2011 has a better initial LT and LO as well as a better radiation hardness as compared to BGO samples grown early. Its light response uniformity is more or less kept up to 5,444 rad/h, indicating possible applications in a severe radiation environment expected at future HEP experiments where fast counting rate is not required. Radiation damage in BGO crystals recovers under room temperature, causing a dose rate dependent damage level. Because of the recovery, a precision light monitoring system is mandatory for a BGO crystal calorimeter in a radiation environment to track variations of crystal's transparency and to provide crucial monitoring corrections for inter-calibration *in situ*.

ACKNOWLEDGMENT

The authors would like to thank colleagues of the SuperB ECAL collaboration. The work presented in this report was inspired by the lively discussion in the SuperB ECAL collaboration.

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